1) a)

$e^{\frac{\pi}{2}i} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$

1) b)

$e^{\frac{3\pi}{4}i} = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$

1) c)

$e^{\pi i} = \cos\pi + i\sin\pi = -1$

1) d)

$e^{\frac{3\pi}{2}i} = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} = -i$

1) e)

$e^{\frac{7\pi}{6}i} = \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6} = -\frac{\sqrt{3}}{2} - \frac{1}{2}i$

1) f)

$e^{2\pi i} = \cos 2\pi + i\sin 2\pi = 1$

2) a)

$z^3 = r^3(\cos\theta + i\sin\theta)^3 = r^3(\cos 3\theta + i\sin 3\theta)$

But $r^3 = 1 \Rightarrow r = 1$

Since $3\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{3}$

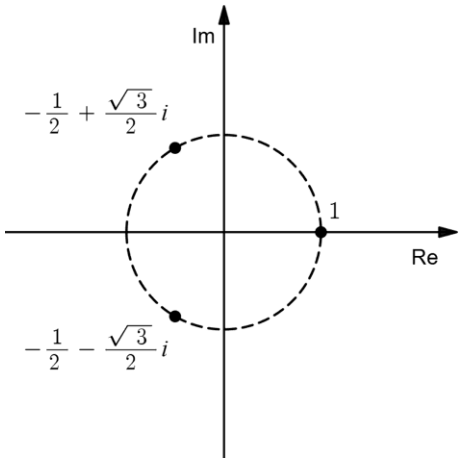
When $k = 0$: $z_0 = e^0 = 1$

When $k = 1$: $z_1 = e^{\frac{2\pi}{3}i} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$

When $k = 2$: $z_2 = e^{\frac{4\pi}{3}i} = e^{-\frac{2\pi}{3}i} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

$\therefore z = 1$, $z = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $z = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

2) b)



2) c)

Equilateral triangle.

2) d)

$z_0 + z_1 + z_2 = 1 - \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} - \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} = 0$

2) e)

Consider a triangle formed with two of the points and the centre:

$\frac{1}{2} \times 1 \times 1 \times \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{4}$

Three triangles form the full shape:

$\frac{3\sqrt{3}}{4}$ units^2

3) a)

$z^5 = r^5(\cos\theta + \mathrm{i}\sin\theta)^5 = r^5(\cos 5\theta + \mathrm{i}\sin 5\theta)$

But $r^5 = 1 \Rightarrow r = 1$

Since $5\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{5}$

When $k = 0$: $z_0 = \mathrm{e}^0$

When $k = 1$: $z_1 = \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}$

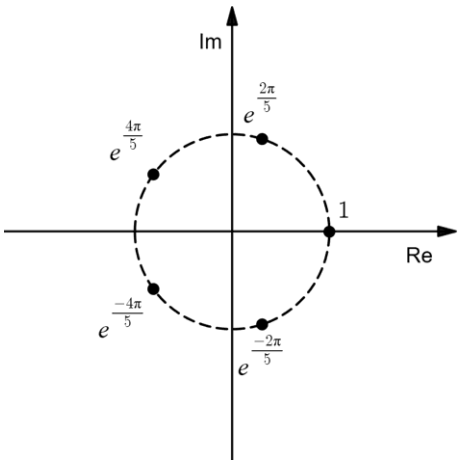
When $k = 2$: $z_2 = \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}$

When $k = 3$: $z_3 = \mathrm{e}^{\frac{6\pi}{5}\mathrm{i}} = \mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}$

When $k = 4$: $z_4 = \mathrm{e}^{\frac{8\pi}{5}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{5}\mathrm{i}}$

$\therefore z = \mathrm{e}^0$, $z = \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}$, $z = \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}$, $z = \mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}$ and $z = \mathrm{e}^{-\frac{2\pi}{5}\mathrm{i}}$

3) b)



3) c)

Regular pentagon.

3) d)

$z_0 + z_1 + z_2 + z_3 + z_4 + z_5 = \mathrm{e}^0 + \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}} + \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}} + \mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}} + \mathrm{e}^{-\frac{2\pi}{5}\mathrm{i}} = 0$

3) e)

Consider a triangle formed with two of the points and the centre:

$\frac{1}{2} \times 1 \times 1 \times \sin\frac{2\pi}{5} = \frac{1}{2}\sin\frac{2\pi}{5}$

Five triangles form the full shape:

$\frac{5}{2}\sin\frac{2\pi}{5} = 2.38$ units^2 (3 SF)

4) a)

$z^6 = r^6(\cos\theta + \mathrm{i}\sin\theta)^6 = r^6(\cos 5\theta + \mathrm{i}\sin 5\theta)$

But $r^6 = 1 \Rightarrow r = 1$

Since $6\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{6}$

When $k = 0$: $z_0 = \mathrm{e}^0 = 1$

When $k = 1$: $z_1 = \mathrm{e}^{\frac{2\pi}{6}\mathrm{i}} = \mathrm{e}^{\frac{\pi}{3}\mathrm{i}} = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$

When $k = 2$: $z_2 = \mathrm{e}^{\frac{4\pi}{6}\mathrm{i}} = \mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$

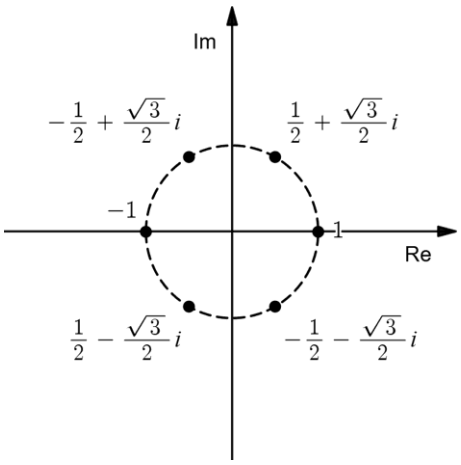
When $k = 3$: $z_3 = \mathrm{e}^{\frac{6\pi}{6}\mathrm{i}} = \mathrm{e}^{\pi\mathrm{i}} = -1$

When $k = 4$: $z_4 = \mathrm{e}^{\frac{8\pi}{6}\mathrm{i}} = \mathrm{e}^{-\frac{\pi}{3}\mathrm{i}} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

When $k = 5$: $z_5 = \mathrm{e}^{\frac{10\pi}{6}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}} = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

$\therefore z = 1$, $z = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, $z = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, $z = -1$, $z = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$ and $z = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

4) b)



4) c)

$z_0 + z_1 + z_2 + z_3 + z_4 + z_5 = 1 + \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} - \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} - 1 + \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} - \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} = 0$

4) d)

Consider a triangle formed with two of the points and the centre:

$\frac{1}{2} \times 1 \times 1 \times \sin\frac{\pi}{3} = \frac{\sqrt{3}}{4}$

Six triangles form the full shape:

$6 \times \frac{\sqrt{3}}{4} = \frac{3\sqrt{3}}{2}$ units^2 (3 SF)

5) a)

$z^7 = r^7(\cos\theta + \mathrm{i}\sin\theta)^7 = r^7(\cos 7\theta + \mathrm{i}\sin 7\theta)$

But $r^7 = 1 \Rightarrow r = 1$

Since $7\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{7}$

When $k = 0$: $z_0 = \mathrm{e}^0$

When $k = 1$: $z_1 = \mathrm{e}^{\frac{2\pi}{7}\mathrm{i}}$

When $k = 2$: $z_2 = \mathrm{e}^{\frac{4\pi}{7}\mathrm{i}}$

When $k = 3$: $z_3 = \mathrm{e}^{\frac{6\pi}{7}\mathrm{i}}$

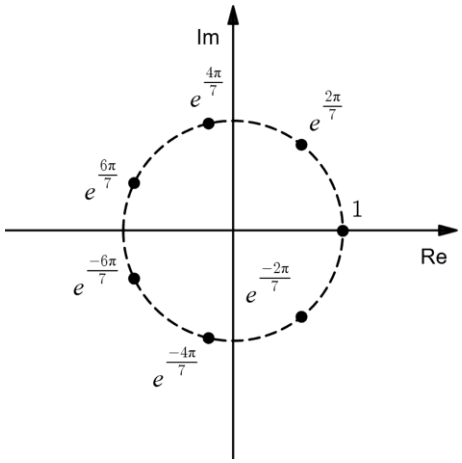
When $k = 4$: $z_4 = \mathrm{e}^{\frac{8\pi}{7}\mathrm{i}} = \mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}}$

When $k = 5$: $z_5 = \mathrm{e}^{\frac{10\pi}{7}\mathrm{i}} = \mathrm{e}^{-\frac{4\pi}{7}\mathrm{i}}$

When $k = 6$: $z_6 = \mathrm{e}^{\frac{12\pi}{7}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{7}\mathrm{i}}$

$\therefore z = \mathrm{e}^0$, $z = \mathrm{e}^{\frac{2\pi}{7}\mathrm{i}}$, $z = \mathrm{e}^{\frac{4\pi}{7}\mathrm{i}}$, $z = \mathrm{e}^{\frac{6\pi}{7}\mathrm{i}}$, $z = \mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}}$, $z = \mathrm{e}^{-\frac{4\pi}{7}\mathrm{i}}$ and $z = \mathrm{e}^{-\frac{2\pi}{7}\mathrm{i}}$

5) b)



5) c)

$e^{0} + e^{\frac{2\pi}{7}i} + e^{\frac{4\pi}{7}i} + e^{\frac{6\pi}{7}i} + e^{-\frac{6\pi}{7}i} + e^{-\frac{4\pi}{7}i} + e^{-\frac{2\pi}{7}i} = 0$

5) d)

Consider a triangle formed with two of the points and the centre:

$\frac{1}{2} \times 1 \times 1 \times \sin\frac{2\pi}{7} = \frac{1}{2}\sin\frac{2\pi}{7}$

Seven triangles form the full shape:

$\frac{7}{2}\sin\frac{2\pi}{7} = 2.74$ units^2 (3 SF)

6) a)

Consider a triangle formed with two of the points and the centre:

$\frac{1}{2} \times 1 \times 1 \times \sin\frac{2\pi}{n} = \frac{1}{2}\sin\frac{2\pi}{n}$

n triangles form the full shape:

$\frac{n}{2}\sin\frac{2\pi}{n}$ units^2

6) b)

As $n \to \infty$ the shape formed gets closer to a circle with radius 1

$\therefore \pi \times 1^2 = \pi$ units^2

7)

Let two consecutive roots of nth roots of unity be $z_1 = \mathrm{e}^{\mathrm{i}\theta}$ and $z_2 = \mathrm{e}^{\mathrm{i}\left(\frac{\theta+2\pi}{n}\right)}$

The angle between them is $\theta = \frac{2\pi}{n}$

The distance between them is the modulus of their difference:

$$|z_2 - z_1| = \left|\mathrm{e}^{\mathrm{i}\left(\frac{\theta+2\pi}{n}\right)} - \mathrm{e}^{\mathrm{i}\theta}\right| = \left|\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{\mathrm{i}\left(\frac{2\pi}{n}\right)} - \mathrm{e}^{\mathrm{i}\theta}\right| = \left|\mathrm{e}^{\mathrm{i}\left(\frac{\theta+\pi}{n}\right)}\right|\left|\mathrm{e}^{\mathrm{i}\left(\frac{\pi}{n}\right)} - \mathrm{e}^{-\mathrm{i}\left(\frac{\pi}{n}\right)}\right| = \left|\mathrm{e}^{\mathrm{i}\left(\frac{\pi}{n}\right)} - \mathrm{e}^{-\mathrm{i}\left(\frac{\pi}{n}\right)}\right|$$

Since $\mathrm{e}^{\mathrm{i}\alpha} - \mathrm{e}^{-\mathrm{i}\alpha} = 2\mathrm{i}\sin\alpha$ then

$$\left|\mathrm{e}^{\mathrm{i}\left(\frac{\pi}{n}\right)} - \mathrm{e}^{-\mathrm{i}\left(\frac{\pi}{n}\right)}\right| = \left|2\mathrm{i}\sin\frac{\pi}{n}\right| = 2\left|\sin\frac{\pi}{n}\right|$$

For there to exist a distance, $n \geq 2$ and so $0 < \frac{\pi}{n} \leq \frac{\pi}{2}$ so that $\sin\frac{\pi}{n} \geq 0$ so

$$|z_2 - z_1| = 2\sin\frac{\pi}{n}$$

Alternatively:

Let two consecutive roots of nth roots of unity be $z_1 = \mathrm{e}^{\mathrm{i}\theta}$ and $z_2 = \mathrm{e}^{\mathrm{i}\left(\frac{\theta+2\pi}{n}\right)}$

The angle between them is $\theta = \frac{2\pi}{n}$

Using the cosine rule the distance is:

$$\sqrt{1^2 + 1^2 - 2 \times 1 \times 1 \times \cos\left(\frac{2\pi}{n}\right)}$$

$$= \sqrt{2 - 2\cos\left(\frac{2\pi}{n}\right)}$$

But since $\cos(2A) = 1 - 2\sin^2(A) \Rightarrow 4\sin^2(A) = 2 - 2\cos(2A)$ which gives

$\sqrt{4\sin^2\left(\frac{\pi}{n}\right)} = \sin\frac{\pi}{n}$ as req.

8) a)

First point: $2\mathrm{e}^0$

Other points: $2\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}}$, $2\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}$

$1, -1 + \sqrt{3}\mathrm{i}$ and $-1 - \sqrt{3}\mathrm{i}$

$\left(-1, \sqrt{3}\right)$ and $\left(-1, -\sqrt{3}\right)$

8) b)

3 triangles of angle $\frac{2\pi}{3}$

$3 \times \left(\frac{1}{2} \times 2 \times 2 \times \sin\frac{2\pi}{3}\right) = 3\sqrt{3}$ units^2

9) a)

First point: $3e^0$

Other points: $3e^{\frac{2\pi}{6}i}$, $3e^{\frac{4\pi}{6}i}$, $3e^{\pi i}$, $3e^{-\frac{4\pi}{6}i}$ and $3e^{-\frac{2\pi}{6}i}$

$\frac{3}{2} + \frac{3\sqrt{3}}{2}i$, $-\frac{3}{2} + \frac{3\sqrt{3}}{2}i$, -3, $-\frac{3}{2} - \frac{3\sqrt{3}}{2}i$ and $\frac{3}{2} - \frac{3\sqrt{3}}{2}i$

$\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right)$, $\left(-\frac{3}{2}, \frac{3\sqrt{3}}{2}\right)$, $(-3, 0)$, $\left(-\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$ and $\left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}i\right)$

9) b)

6 triangles of angle $\frac{\pi}{3}$

$6 \times \left(\frac{1}{2} \times 3 \times 3 \times \sin\frac{\pi}{3}\right) = \frac{27\sqrt{3}}{2}$ units^2

10)

Let the root be $z = e^{i\theta}$ and so its complex conjugate is $z^* = e^{-i\theta}$

The nth power of the conjugate is: $(z^*)^n = \left(e^{-i\theta}\right)^n = e^{-in\theta}$

Since $z^n = e^{in\theta} = 1$ then

$(z^*)^n = e^{-in\theta} = \left(e^{in\theta}\right)^* = (1)^* = 1$

So z^* is also an nth root of unity.

11)

Assume the first root is at 1, so if there is only one root then it is true.

Let $\omega = e^{\frac{2\pi}{n}i}$ so that the sum of the roots is:

$$S_n = 1 + \omega + \omega^2 + \cdots \omega^{n-1}$$

This is a geometric series with $a = 1$, $r = \omega$ and n terms:

$S_n = \frac{a(1-r^n)}{1-r} = \frac{1(1-\omega^n)}{1-\omega}$

But since $\omega^n = 1$

$S_n = \frac{1(1-1)}{1-\omega} = 0$

12) a)

$\left(\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}\right)^5=\cos\left(5\times\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(5\times\frac{4\pi}{5}\right)=1+0\mathrm{i}=1$ so $\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}$ is a solution to $z^5=1$

12) b)

$\left(\mathrm{e}^{\frac{2\pi}{7}\mathrm{i}}\right)^7=\mathrm{e}^{7\times\frac{2\pi}{7}\mathrm{i}}=\mathrm{e}^{2\pi\mathrm{i}}=\mathrm{e}^0=1$ so $\mathrm{e}^{\frac{2\pi}{7}\mathrm{i}}$ is a root of the equation $z^7=1$

13) a)

$z^5=r^5(\cos\theta+\mathrm{i}\sin\theta)^5=r^5(\cos 5\theta+\mathrm{i}\sin 5\theta)$

But $r^5=1\Rightarrow r=1$

Since $5\theta=0+2k\pi$ for integer k: $\theta=\frac{2k\pi}{5}$

When $k=0$: $z_0=\cos 0+\mathrm{i}\sin 0$

When $k=1$: $z_1=\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}$

When $k=2$: $z_2=\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}$

When $k=3$: $z_3=\cos\frac{6\pi}{5}+\mathrm{i}\sin\frac{6\pi}{5}=\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right)$

When $k=4$: $z_4=\cos\frac{8\pi}{5}+\mathrm{i}\sin\frac{8\pi}{5}=\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right)$

$\therefore z=\cos 0+\mathrm{i}\sin 0$, $z=\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}$, $z=\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}$,

$z=\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right)$ and $z=\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right)$

13) b)

$\cos 0+\mathrm{i}\sin 0+\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}+\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}+\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right)+$

$\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right)=0$

Consider the real parts:

$1+\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}+\cos\left(-\frac{4\pi}{5}\right)+\cos\left(-\frac{2\pi}{5}\right)=0$

Since $\cos\frac{2\pi}{5}=\cos\left(-\frac{2\pi}{5}\right)$ and $\cos\frac{4\pi}{5}=\cos\left(-\frac{4\pi}{5}\right)$

$1+2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=0$

$\Rightarrow 2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=-1$

$\therefore\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}=-\frac{1}{2}$ as required.

14) a)

$z^7 = r^7(\cos\theta + \mathrm{i}\sin\theta)^7 = r^7(\cos 7\theta + \mathrm{i}\sin 7\theta)$

But $r^7 = 1 \Rightarrow r = 1$

Since $7\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{7}$

When $k = 0$: $z_0 = \cos 0 + \mathrm{i}\sin 0$

When $k = 1$: $z_1 = \cos\frac{2\pi}{7} + \mathrm{i}\sin\frac{2\pi}{7}$

When $k = 2$: $z_2 = \cos\frac{4\pi}{7} + \mathrm{i}\sin\frac{4\pi}{7}$

When $k = 3$: $z_3 = \cos\frac{6\pi}{7} + \mathrm{i}\sin\frac{6\pi}{7}$

When $k = 4$: $z_4 = \cos\frac{8\pi}{7} + \mathrm{i}\sin\frac{8\pi}{7} = \cos\left(-\frac{6\pi}{7}\right) + \mathrm{i}\sin\left(-\frac{6\pi}{7}\right)$

When $k = 5$: $z_5 = \cos\frac{10\pi}{7} + \mathrm{i}\sin\frac{10\pi}{7} = \cos\left(-\frac{4\pi}{7}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{7}\right)$

When $k = 6$: $z_6 = \cos\frac{12\pi}{7} + \mathrm{i}\sin\frac{12\pi}{7} = \cos\left(-\frac{2\pi}{7}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{7}\right)$

$\therefore z = \cos 0 + \mathrm{i}\sin 0$, $z = \cos\frac{2\pi}{7} + \mathrm{i}\sin\frac{2\pi}{7}$, $z = \cos\frac{4\pi}{7} + \mathrm{i}\sin\frac{4\pi}{7}$, $z = \cos\frac{6\pi}{7} + \mathrm{i}\sin\frac{6\pi}{7}$,

$z = \cos\left(-\frac{6\pi}{7}\right) + \mathrm{i}\sin\left(-\frac{6\pi}{7}\right)$, $z = \cos\left(-\frac{4\pi}{7}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{7}\right)$

and $z = \cos\left(-\frac{2\pi}{7}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{7}\right)$

14) b)

$\cos 0 + \mathrm{i}\sin 0 + \cos\frac{2\pi}{7} + \mathrm{i}\sin\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \mathrm{i}\sin\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \mathrm{i}\sin\frac{6\pi}{7} + \cos\left(-\frac{6\pi}{7}\right) +$

$\mathrm{i}\sin\left(-\frac{6\pi}{7}\right) + \cos\left(-\frac{4\pi}{7}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{7}\right) + \cos\left(-\frac{2\pi}{7}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{7}\right) = 0$

Consider the real parts:

$1 + \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \cos\left(-\frac{6\pi}{7}\right) + \cos\left(-\frac{4\pi}{7}\right) + \cos\left(-\frac{2\pi}{7}\right) = 0$

Since $\cos\frac{2\pi}{7} = \cos\left(-\frac{2\pi}{7}\right)$, $\cos\frac{4\pi}{7} = \cos\left(-\frac{4\pi}{7}\right)$ and $\cos\frac{6\pi}{7} = \cos\left(-\frac{6\pi}{7}\right)$

$1 + 2\cos\frac{2\pi}{7} + 2\cos\frac{4\pi}{7} + 2\cos\frac{6\pi}{7} = 0$

$\Rightarrow 2\cos\frac{2\pi}{7} + 2\cos\frac{4\pi}{7} + 2\cos\frac{6\pi}{7} = -1$

$\therefore \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} = -\frac{1}{2}$ as required.

15)

The five roots of unity are

$\cos 0 + \mathrm{i}\sin 0$, $\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}$, $\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}$, $\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)$ and $\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)$

$$\cos 0 + \mathrm{i}\sin 0 + \cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5} + \cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5} + \cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right) + \cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right) = 0$$

Consider the real parts:

$$1 + \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} + \cos\left(-\frac{4\pi}{5}\right) + \cos\left(-\frac{2\pi}{5}\right) = 0$$

Since $\cos a = \cos(-a)$ then

$$1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0 \Rightarrow \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2} \Rightarrow -\cos\frac{2\pi}{5} - \cos\frac{4\pi}{5} = \frac{1}{2}$$

However, we know that $\cos(\pi - \alpha) = -\cos\alpha$:

Since $\cos\frac{3\pi}{5} = \cos\left(\pi - \frac{2\pi}{5}\right) = -\cos\frac{2\pi}{5}$ and $\cos\frac{\pi}{5} = \cos\left(\pi - \frac{4\pi}{5}\right) = -\cos\frac{4\pi}{5}$

$\therefore \cos\frac{3\pi}{5} + \cos\frac{\pi}{5} = \frac{1}{2}$ as req.

Alternatively:

The five roots of unity are

$\cos 0 + \mathrm{i}\sin 0$, $\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}$, $\cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}$, $\cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right)$ and $\cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right)$

$$\cos 0 + \mathrm{i}\sin 0 + \cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5} + \cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5} + \cos\left(-\frac{4\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{4\pi}{5}\right) + \cos\left(-\frac{2\pi}{5}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{5}\right) = 0$$

Consider the real parts:

$$1 + \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} + \cos\left(-\frac{4\pi}{5}\right) + \cos\left(-\frac{2\pi}{5}\right) = 0$$

Since $\cos a = \cos(-a)$ then

$$1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0 \Rightarrow \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2} \Rightarrow -\cos\frac{2\pi}{5} - \cos\frac{4\pi}{5} = \frac{1}{2}$$

However, we know that $\cos\alpha = -(\cos\alpha - \pi)$:

$$\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}$$

$$-\cos\left(\frac{2\pi}{5} - \pi\right) - \cos\left(\frac{4\pi}{5} - \pi\right) = -\frac{1}{2}$$

$$-\cos\left(-\frac{3\pi}{5}\right) - \cos\left(-\frac{\pi}{5}\right) = -\frac{1}{2}$$

$\therefore \cos\frac{3\pi}{5} + \cos\frac{\pi}{5} = \frac{1}{2}$ as req.

16) a)

Since $3\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{3}$

When $k = 0$: $z_0 = \mathrm{e}^0 = 1$

When $k = 1$: $z_1 = \mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$

When $k = 2$: $z_2 = \mathrm{e}^{\frac{4\pi}{3}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

$\therefore 1, -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

$$\left(z - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z - \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)\right) = z^2 + \left(\frac{1}{2} + \frac{1}{2}\right)z + \left(\frac{1}{4} + \frac{3}{4}\right) = z^2 + z + 1$$

So $(z-1)(z^2 + z + 1) = z^3 + z^2 + z - z^2 - z - 1 = z^3 - 1$

16) b)

From part b) the three cube roots of unity are $1, -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

So the midpoints between each pair on the Argand diagram are:

1 and $-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ gives: $\frac{1}{4} + \frac{\sqrt{3}}{4}\mathrm{i}$

$-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$ gives: $-\frac{1}{2}$

$-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$ and 1 gives: $\frac{1}{4} - \frac{\sqrt{3}}{4}\mathrm{i}$

$$\left(z - \left(\frac{1}{4} + \frac{\sqrt{3}}{4}\mathrm{i}\right)\right)\left(z - \left(\frac{1}{4} - \frac{\sqrt{3}}{4}\mathrm{i}\right)\right) = z^2 + \left(-\frac{1}{4} - \frac{1}{4}\right)z + \left(\frac{1}{16} + \frac{3}{16}\right) = z^2 - \frac{1}{2}z + \frac{1}{4}$$

$$\left(z^2 - \frac{1}{2}z + \frac{1}{4}\right)\left(z + \frac{1}{2}\right) = z^3 - \frac{1}{2}z^2 + \frac{1}{4}z + \frac{1}{2}z^2 - \frac{1}{4}z + \frac{1}{8} = z^3 + \frac{1}{8}$$

So since $z^3 + \frac{1}{8} = 0$ then $8z^3 + 1 = 0$

$\therefore 8z^3 + 1$

17) a)

$1 + z + z^2 + z^3 + z^4 + z^5$ is a geometric series so we are solving $\frac{z^6-1}{z-1} = 0$

Since we are using $\frac{z^6-1}{z-1} = 0$ we know that $z \neq 1$

So $1 + z + z^2 + z^3 + z^4 + z^5 = 0$ has solutions of the 6th roots of unity:

The six roots of unity are found as follows:

$6\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{6}$

When $k = 0$: $z_0 = \mathrm{e}^0 = 1$

When $k = 1$: $z_1 = \mathrm{e}^{\frac{2\pi}{6}\mathrm{i}} = \mathrm{e}^{\frac{\pi}{3}\mathrm{i}} = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$

When $k = 2$: $z_2 = \mathrm{e}^{\frac{4\pi}{6}\mathrm{i}} = \mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$

When $k = 3$: $z_3 = \mathrm{e}^{\frac{6\pi}{6}\mathrm{i}} = \mathrm{e}^{\pi\mathrm{i}} = -1$

When $k = 4$: $z_4 = \mathrm{e}^{\frac{8\pi}{6}\mathrm{i}} = \mathrm{e}^{-\frac{\pi}{3}\mathrm{i}} = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

When $k = 5$: $z_5 = \mathrm{e}^{\frac{8\pi}{6}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

$\therefore$ the solutions to $1 + z + z^2 + z^3 + z^4 + z^5 = 0$ are

$z = -1$, $z = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$, $z = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$, $z = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $z = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$

17) b)

$$\left(z - \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right)\right) = z^2 + \left(\frac{1}{2} + \frac{1}{2}\right)z + \left(\frac{1}{4}\right) - \left(-\frac{3}{4}\right) = z^2 + z + 1$$

$$\left(z - \left(\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z - \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right)\right) == z^2 - \left(\frac{1}{2} + \frac{1}{2}\right)z + \left(\frac{1}{4}\right) - \left(-\frac{3}{4}\right) = z^2 - z + 1$$

$\therefore 1 + z + z^2 + z^3 + z^4 + z^5 = (z + 1)(z^2 + z + 1)(z^2 - z + 1)$

18) a)

$1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7$ is a geometric series so we are solving $\frac{z^7-1}{z-1} = 0$

Since we are using $\frac{z^7-1}{z-1} = 0$ we know that $z \neq 1$

So $1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7 = 0$ has solutions of the 8th roots of unity:

The eight roots of unity are found as follows:

$8\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{4}$

When $k = 0$: $z_0 = \mathrm{e}^0 = 1$

When $k = 1$: $z_1 = \mathrm{e}^{\frac{2\pi}{8}\mathrm{i}} = \mathrm{e}^{\frac{\pi}{4}\mathrm{i}} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$

When $k = 2$: $z_2 = \mathrm{e}^{\frac{4\pi}{8}\mathrm{i}} = \mathrm{e}^{\frac{\pi}{2}\mathrm{i}} = \mathrm{i}$

When $k = 3$: $z_3 = \mathrm{e}^{\frac{6\pi}{8}\mathrm{i}} = \mathrm{e}^{\frac{3\pi}{4}\mathrm{i}} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$

When $k = 4$: $z_4 = \mathrm{e}^{\frac{8\pi}{8}\mathrm{i}} = \mathrm{e}^{\pi\mathrm{i}} = -1$

When $k = 5$: $z_5 = \mathrm{e}^{\frac{10\pi}{8}\mathrm{i}} = \mathrm{e}^{-\frac{3\pi}{4}\mathrm{i}} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}$

When $k = 6$: $z_6 = \mathrm{e}^{\frac{12\pi}{8}\mathrm{i}} = \mathrm{e}^{-\frac{\pi}{3}\mathrm{i}} = -\mathrm{i}$

When $k = 7$: $z_7 = \mathrm{e}^{\frac{14\pi}{8}\mathrm{i}} = \mathrm{e}^{-\frac{\pi}{2}\mathrm{i}} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}$

$\therefore$ the solutions to $1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7 = 0$ are

$z = -1$, $z = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}$, $z = -\mathrm{i}$, $z = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}$, $z = 1$, $z = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$, $z = \mathrm{i}$

and $z = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$

18) b)

$$\left(z - \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}\right)\right)\left(z - \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}\right)\right) = z^2 - \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right)z + \left(\frac{2}{4} + \frac{2}{4}\right) = z^2 - \sqrt{2}z + 1$$

$$(z - \mathrm{i})\big(z - (-\mathrm{i})\big) = z^2 + 1$$

$$\left(z - \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}\right)\right)\left(z - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}\right)\right) = z^2 + \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right)z + \left(\frac{2}{4} + \frac{2}{4}\right) = z^2 + \sqrt{2}z + 1$$

And

$$\big(z^2 - \sqrt{2}z + 1\big)\big(z^2 + \sqrt{2}z + 1\big)$$

$$= z^4 + \big(\sqrt{2} - \sqrt{2}\big)z^3 + (1 + 1 - 2)z^2 + \big(-\sqrt{2} + \sqrt{2}\big)z + 1 = (z^4 + 1)$$

$$\therefore 1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7 = (z + 1)(z^2 + 1)(z^4 + 1)$$

19) a)

Roots are 1, ω, ω^2 so $\omega^3 = 1$

19) b)

Roots are 1, ω, ω^2 so $\omega^5 = \omega^2$

19) c)

Roots are 1, ω, ω^2 so

$\omega^8 = \omega^5 = \omega^2$

$\omega^6 = \omega^3 = 1$

$\therefore \omega^8 - \omega^5 + \omega^6 = \omega^2 - \omega^2 + 1 = 1$

19) d)

Roots are 1, ω, ω^2 so

$\frac{1}{\omega} = \omega^{-1} = \omega^2$

20) a)

$\omega = \omega^1 = \mathrm{e}^{\frac{2\pi}{n}\mathrm{i}}$ so $\omega^* = \mathrm{e}^{-\frac{2\pi}{n}\mathrm{i}}$

But since they are roots of unity: $\omega^* = \mathrm{e}^{-\frac{2\pi}{n}\mathrm{i}} = \mathrm{e}^{\frac{2(n-1)\pi}{n}\mathrm{i}} = \omega^{n-1}$

Alternatively:

$\omega^n = 1$ So $\omega \times \omega^* = 1$

$\Rightarrow \omega \times \omega^* = \omega^n \therefore \omega^* = \omega^{n-1}$

20) b)

$(\omega^k)^* = (\omega^*)^k = (\omega^{n-1})^k = \omega^{k(n-1)}$

Which given it has n roots gives $(\omega^k)^* = \omega^{n-k}$

Alternatively:

$(\omega^k)^n = 1$ So $\omega^k \times (\omega^k)^* = 1$

$\Rightarrow \omega^k \times (\omega^k)^* = (\omega^k)^n \therefore (\omega^k)^* = \omega^{k(n-1)}$

21) a)

Six roots of unity so $1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 = 0$

$\mathrm{Re}(\omega + \omega^2 + \omega^3 + \omega^4 + \omega^5) = \mathrm{Re}(-1) = -1$

21) b)

Since $\omega = \mathrm{e}^{\frac{2\pi}{6}\mathrm{i}}$ then $\omega^2 = \mathrm{e}^{\frac{2\pi}{3}\mathrm{i}}$ and $\omega^4 = \mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}$

$\mathrm{Im}(\omega^2 + \omega^4) = \mathrm{Im}\left(\mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} + \mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}\right) = \sin\left(\frac{2\pi}{3}\right) + \sin\left(-\frac{2\pi}{3}\right) = 0$

22) a)

Since $\omega = \mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$ then $\omega^2 = \mathrm{e}^{\frac{2\pi}{4}\mathrm{i}}$

$\mathrm{Re}(\omega^2 + \omega) = \cos\frac{\pi}{2} + \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$

22) b)

$\omega^4 = 0$

$\mathrm{Im}(1 + \omega^4) = \mathrm{Im}(1 + 0) = 0$

23)

The four roots of unity are 1, $\omega = \mathrm{i}$, $\omega^2 = -1$, $\omega^3 = -\mathrm{i}$

$(1 - \mathrm{i})(1 + 1)(1 + \mathrm{i}) = 2(1 + \mathrm{i})(1 - \mathrm{i})$

$= 2(1 + 1) = 4$ as req.

24) a)

$a = 1, r = x, n = n$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{1(1-x^n)}{1-x} = \frac{1-x^n}{1-x}$

24) b)

Since $z^n - 1 = (z-1)(z-\omega)(z-\omega^2)\ldots(z-\omega^{n-1})$

$\therefore$ as $z \neq 1$ then can divide by $z-1$:

$\frac{z^n-1}{z-1} = (z-\omega)(z-\omega^2)\ldots(z-\omega^{n-1})$ which from part a) is $1 + z + z^2 + \cdots + z^{n-1}$

Since both sides are equal for all $z \neq 1$, and both define the same polynomial for $z \neq 1$, now take the limit as $z \to 1$:

in $1 + z + z^2 + \cdots + z^{n-1}$ gives $1 + 1 + 1 + \cdots + 1 = n$

and in $(z-\omega)(z-\omega^2)\ldots(z-\omega^{n-1})$ gives $(1-\omega)(1-\omega^2)\ldots(1-\omega^{n-1})$

So that

$$\prod_{k=1}^{n-1}(1-\omega^k) = n$$

as required.

25) a)

Since $|z|^2 = zz^*$ and let $z = z_1 + z_2$:

$|z_1 + z_2|^2 = (z_1 + z_2)(z_1 + z_2)^*$

Since conjugation distributes over addition:

$|z_1 + z_2|^2 = (z_1 + z_2)(z_1^* + z_2^*)$

25) b)

$|\omega - \omega^2|^2 = (\omega - \omega^2)(\omega^* - (\omega^2)^*)$

Since $(\omega^2)^* = (\omega^*)^2$:

$|\omega - \omega^2|^2 = (\omega - \omega^2)(\omega^* - (\omega^*)^2)$

$= \omega\omega^* - \omega(\omega^*)^2 - \omega^2\omega^* + \omega^2(\omega^*)^2$

Since $|\omega|^2 = \omega\omega^* = 1$ then

$= 1 - \omega(\omega^*)^2 - \omega^2\omega^* + 1$

$= 2 - (\omega(\omega^*)^2 + \omega^2\omega^*)$

Since ω is a cube root of unity, $\omega^2\omega^* = \frac{\omega^2}{\omega} = \omega$ and $\omega(\omega^*)^2 = \frac{\omega}{\omega^2} = \frac{1}{\omega} = \omega^*$

$\therefore |\omega - \omega^2|^2 = 2 - (\omega + \omega^*)$ as req.

Alternatively:

For $\omega = 1$:

LHS: $|1 - 1^2|^2 = 0$ RHS: $2 - (1 + 1^*) = 0$ $\therefore$ LHS = RHS so true for $\omega = 1$

For $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$:

LHS: $\left|-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right)^2\right|^2 = \left|-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} + \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right|^2 = \left|\sqrt{3}\mathrm{i}\right|^2 = 3$

RHS: $2 - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} - \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right) = 3$

$\therefore$ LHS = RHS so true for $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$

For $\omega = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$:

LHS: $\left|-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} - \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)^2\right|^2 = \left|-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} + \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right|^2 = \left|-\sqrt{3}\mathrm{i}\right|^2 = 3$

RHS: $2 - \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} - \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right) = 3$

$\therefore$ LHS = RHS so true for $\omega = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

$\therefore |\omega - \omega^2|^2 = 2 - (\omega + \omega^*)$ as req.

26) a)

Since $1 + \omega + \omega^2 = 0$:

$(1 - \omega)(1 - \omega^2) = 1 - (\omega + \omega^2) + \omega^3 = 1 - (-1) + 1$

$= 3$

26) b)

$1 - \omega^3 = 1 - 1 = 0$ and from part a) $(1 - \omega)(1 - \omega^2) = 3$

$\therefore\ (1 - \omega)(1 - \omega^2)(1 - \omega^3) = 3 \times 0 = 0$

26) c)

Since $1 + \omega + \omega^2 = 0$:

$(1 + \omega)(1 + \omega^2) = -\omega^2 \times -\omega = \omega^3 = 1$

26) d)

$\omega^3 = 1$ and from part c) $(1 + \omega)(1 + \omega^2) = 1$ so

$(1 + \omega)(1 + \omega^2)(1 + \omega^3) = 1 \times (1 + 1) = 2$

Alternatively:

Since $1 + \omega + \omega^2 = 0$:

$(1 + \omega)(1 + \omega^2)(1 + \omega^3) = -\omega^2 \times -\omega \times 2 = 2\omega^3 = 2$

27) a)

Since $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$

$(a - b\omega)(a - b\omega^2) = a^2 - ab(\omega + \omega^2) + b^2\omega^3 = a^2 - ab(-1) + b^2 = a^2 + ab + b^2$

27) b)

$\omega^3 = 1$ and from part a) $(a - b\omega)(a - b\omega^2) = a^2 + ab + b^2$

$\therefore (a - b\omega)(a - b\omega^2)(a - b\omega^3)$

$= (a^2 + ab + b^2)(a - b)$

$= a^3 - b^3$

27) c)

Since $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$

$(a + b\omega)(a + b\omega^2) = a^2 + ab(\omega + \omega^2) + b^2\omega^3 = a^2 - ab + b^2$

27) d)

Since $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$

$(a + b\omega)(a + b\omega^2)(a + b\omega^3)$

We saw $(a + b\omega)(a + b\omega^2)$ in part c) so

$(a^2 - ab + b^2)(a + b\omega^3)$

$= (a^2 - ab + b^2)(a + b)$

$= a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$

$= a^3 + b^3$

28) a)

Four roots of unity tells us $1 + \omega + \omega^2 + \omega^3 = 0$

$\therefore \omega(1 + \omega + \omega^2) = \omega \times -\omega^3 = -\omega^4 = -1$

28) b)

Four roots of unity tells us $1 + \omega + \omega^2 + \omega^3 = 0$

$\therefore \omega(1 + \omega)(1 + \omega^2) = \omega(1 + \omega + \omega^2 + \omega^3) = 0$

28) c)

Four roots of unity tells us $1 + \omega + \omega^2 + \omega^3 = 0$

$\left(\omega + \frac{1}{\omega}\right)^2 + 2\left(\omega + \frac{1}{\omega}\right) = \omega^2 + 2 + \omega^{-2} + 2\omega + 2\omega^{-1}$

But since $\omega^{-2} = \omega^2$ and $\omega^{-1} = \omega^3$

$= \omega^2 + 2 + \omega^2 + 2\omega + 2\omega^3 = 2\omega^2 + 2 + 2\omega + 2\omega^3 = 2(1 + \omega + \omega^2 + \omega^3) = 0$

29) a)

$\omega + \frac{1}{\omega} + 1 = \omega + \omega^{-1} + 1$

But since as there are three roots of unity then $\omega^{-1} = \omega^2$

$= \omega + \omega^2 + 1 = 0$

29) b)

From part a), let the sum of the conjugates be x:

$x = \omega + \frac{1}{\omega}$

So that $x + 1 = 0$

$\therefore x = -1$

Since $\omega = \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}$ and $\omega^* = \frac{1}{\omega}$:

$x = \omega + \frac{1}{\omega} = \left(\cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3}\right) + \left(\cos\frac{2\pi}{3} - \mathrm{i}\sin\frac{2\pi}{3}\right) = 2\cos\frac{2\pi}{3}$

So that $2\cos\frac{2\pi}{3} = -1$

$\therefore \cos\frac{2\pi}{3} = -\frac{1}{2}$ as req.

30) a)

$\left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega + \frac{1}{\omega}\right) = \omega^2 + 2 + \omega^{-2} + \omega + \omega^{-1}$

But since as there are five roots of unity then $\omega^{-2} = \omega^3$ and $\omega^{-1} = \omega^4$

$= \omega^2 + 2 + \omega^3 + \omega + \omega^4 = 1 + 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 1 + 0 = 1$

30) b)

From part a), let the sum of the conjugates be x:

$x = \omega + \frac{1}{\omega}$

So that $x^2 + x = 1 \Rightarrow x^2 + x - 1 = 0$

$\therefore x = \frac{-1 \pm \sqrt{5}}{2}$

Since $\omega = \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}$ and $\omega^* = \frac{1}{\omega}$:

$x = \omega + \frac{1}{\omega} = \omega + \omega^4 = \left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right) + \left(\cos\frac{2\pi}{5} - \mathrm{i}\sin\frac{2\pi}{5}\right) = 2\cos\frac{2\pi}{5}$

But $\omega = \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}$ lies in the first quadrant, so $2\cos\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{2}$

But we want $\cos\frac{4\pi}{5}$:

$\cos\frac{4\pi}{5} = \cos\left(2 \times \frac{2\pi}{5}\right) = 2\left(\cos\frac{2\pi}{5}\right)^2 - 1 = 2\left(\frac{-1+\sqrt{5}}{4}\right)^2 - 1 = \frac{-1-\sqrt{5}}{4}$

$\therefore \cos\frac{4\pi}{5} = \frac{-1-\sqrt{5}}{4}$